

Collections Recovery: What Actually Happened

Executive Memo — Data Analyst Assignment | Golden dataset, Jan–Jul 2026 (7 complete months)

What happened

Recovery is flat, not improving. Seven complete months of Golden (de-duplicated) recovery data show no statistically detectable trend — a linear regression across Jan–Jul returns $p = 0.89$. The reported “11% month-on-month improvement” is real as arithmetic (March was 11.0% above February) but not real as a signal: February was the low point of the window (−9.1% vs. January), so March’s “improvement” is a rebound to roughly where January already was — March is only +0.9% above January. Every headline metric we reconstructed tells the same story: contact rate (22.5–24.0%), PTP rate (13.9–16.6%), PTP kept rate (48.5–50.7%), and recovery per agent-hour (₹16.1k–₹17.0k) all move within normal month-to-month noise, not trend.

Why it happened

We tested every driver the assignment asked about — agent, agent tenure, telephony vendor, channel, campaign, attempt frequency, portfolio mix, targeting-strategy proxy, and targeting priority — and **none showed an effect distinguishable from statistical noise**. Agent-to-agent variance in answer rate (std 4.3%) matches pure sampling noise (expected 4.2%) almost exactly. All 15 telephony vendors perform identically (19.3–20.5% answer rate). Portfolio mix (risk segment, loan type) is stable within $\pm 3\text{pp}$ every month. As a final, more rigorous check, a **multivariate logistic regression on 89,071 account-months, standard errors clustered by account** — controlling for month, risk segment, loan type, DPD bucket, attempt count, and channel touches simultaneously — confirms the month effects are jointly indistinguishable from zero (cluster-robust Wald $p = 0.659$). A power analysis confirms the test could have caught an effect this size: 80% power to detect a $\pm 0.94\text{pp}$ ($\approx 12\%$ relative) swing, versus $\approx 0.65\text{pp}$ actually observed. Geography and language could not be tested: 77% of borrowers have internally conflicting city/phone/email data, and no language field exists in the source at all.

How confident are we

- **High confidence** the 11% figure is not evidence of a sustained trend — confirmed by simple trend tests and a cluster-robust multivariate regression that a power analysis shows was strong enough to detect an effect of that scale.
- **High confidence** the previously reported recovery figure was inflated — 500 duplicate payment rows overstated recovery by ₹2.59 Cr (1.93%), now corrected in the Golden dataset.
- **Low confidence** in any campaign-level or channel-level attribution claim — 68% of successful payments have no traceable campaign touch within 30 days, and the attribution window choice changes the attributed campaign for 27% of payments that can be attributed at all.
- **Escalated, unresolved:** account status disagrees between the core dimension table and the event log for 85.7% of accounts with history — no data-level way to pick a winner; needs an Engineering decision.

What we should do

Retire the “11% MoM” headline and replace it with a control-chart view of recovery (mean ± 1 std dev) so a normal-noise month is never again reported as a result. Fix the duplicate-payment and disposition-code-synonym issues in the reporting pipeline (both already fixed in the Golden dataset delivered with this analysis). Escalate the accounts.status vs. event-log conflict to Engineering — it affects every denominator-based metric in the business.

For the ₹10 Cr investment: deploy it as a **gated two-phase bet on WhatsApp / digital engagement** — the one lever with even a weak, directional positive signal, and by far the cheapest and most reversible to test. 86% of the book is already WhatsApp-reachable but touched only once every 13 weeks; frequency, not reach, is the binding constraint.

- **Phase 0 (₹1.2 Cr, months 1–3):** a randomized controlled pilot — 3x cadence + AI-personalized sequencing on $\sim 13,000$ accounts vs. a held-out control, pre-registered 14-day outcome window.
- **Phase 1 (₹8.8 Cr, months 4–15):** scaled to the full 30,000-account book — released only if Phase 0 shows a statistically significant, positive lift. If it doesn’t, the loss is capped at 12% of the budget, not all of it.

Expected financial impact

Because the underlying incrementality of digital cadence has not been measured — only weakly, non-significantly suggested — we present a range rather than a single number:

Scenario	Incrementality	Year-1 incremental recovery	ROI on ₹10 Cr
Downside (3%)	3%	₹5.5 Cr	−45%
Base case (8%)	8%	₹14.7 Cr	+47%
Upside (15%)	15%	₹27.5 Cr	+175%

Bottom line: do not commit the full ₹10 Cr on the strength of an untested assumption — that is exactly the mistake behind the “11% MoM” headline this memo corrects. Spend 12% of the budget to find out for certain, then scale only what is proven.